

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November - December 2012

CL 205 Building Materials and Construction

Date: 01.12.2012, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Write composition of cement. What are the requirements of mortar? [04]
(b) Define following terms related to staircase: [03]
(i) Nosing (ii) Balustrade (iii) Soffit
- Q - 2 (a) Write the properties of a good brick. [04]
(b) Give the classification of timber. Write uses of it. [05]
(c) Enlist types of ground floors. Explain flag stone flooring in detail with figure. [05]

OR

- Q - 2 (a) What are uses of ceramics? [04]
(b) Write requirements and uses of paints, varnishes and distempers. [05]
(c) List out types of upper floors. Discuss any one type with sketch. [05]
- Q - 3 (a) Enlist thermal, electrical and magnetic properties of engineering materials. [05]
(b) What are requirements of a good roof? List out types of roof. [05]
(c) Draw a neat sketch of bifurcated stair. [04]

OR

- Q - 3 (a) What are the physical properties of metals, non-metals and alloys? [05]
(b) Enlist types of roof covering. What are the advantages and disadvantages of flat terraced roof? [05]
(c) Draw a neat sketch of dog-legged staircase with well. [04]

SECTION - II

- Q - 4 Differentiate between English & Flemish bond with neat sketches showing elevation. [7]
- Q - 5 (a) Draw neat figures of ashlar fine tooled and chamfered fine tooled masonry works. [2]
(b) What is timbering in trenches? Enlist methods of it. Explain any one of them briefly. [3]

- (c) Design strip footing for a brick wall 30 cm thick and 3.4 m high above the ground level. The wall carries a superimposed load of 90 kN/m run. The soil has unit weight of 17 kN/m³, angle of repose of 29° and safe bearing capacity 140 kN/m². The footing may have lime concrete base with unit weight of 20 kN/m³ and modulus of rupture 100 kN/m². Assume unit weight of masonry is 19.5 kN/m³. [9]

OR

- Q - 5 (a) List out various types of composite masonry. Briefly explain stone facing with brick backing masonry work. [2]
- (b) Define the terms scaffolding, shoring and underpinning. [3]
- (c) Classify foundations by different ways. What are the needs and essential requirements to provide it? [4]
- (d) What are the causes of failure of foundation? What remedial measures should be taken to prevent it? [5]
- Q - 6 (a) Draw neat sketches of following showing all components: [9]
- (i) A typical door frame (ii) Battened, ledged & braced door (iii) Pivoted window
- (b) Define plastering. What are the objectives of it? Enlist various defects observed in plastering. [5]

OR

- Q - 6 (a) Draw neat sketches of following: [9]
- (i) Solid core flush door (ii) Revolving door (iii) Bay window
- (b) List out types of pointing and draw neat sketches of it. [5]
